DisasterMind — Technical & Product Overview

An autonomous, multi-agent disaster-coordination platform with validated, real-data risk prediction and a human-in-the-loop command pipeline.

DisasterMind ingests live hazard signals for cyclones/floods, earthquakes, and urban/forest fires; predicts per-asset risk with models validated on real historical events; models cascading effects; plans resources and evacuation routes; and drives a human-commanded decision pipeline that ends in dispatch — all surfaced through a browser command-and-control console.

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1. Executive summary

What it is	A decision-support system for multi-hazard disaster early warning and coordination.
Hazards	Cyclone/flood, earthquake (rapid impact assessment), urban/forest fire.
Differentiator	Every model is validated on **real historical data**, leak-free, and shown to beat the operational baselines with statistical significance — not a demo on synthetic numbers.
Design stance	Standard-library-first and degrades gracefully; explicitly *decision-support* (a human commander holds authority) with a tamper-evident audit trail.
Scale	~39,000 lines of Python across 34 subsystems, a Vite + React 19 + TypeScript console (74 source modules), and a suite of ~1,110 offline, deterministic Python tests alongside a Vitest console suite.

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2. The problem

Disaster response fails at the seams between sensing, prediction, and action: warnings arrive too late to act on, forecasts are accurate on average but blind to the vulnerable, alerts reach phones but not people, and decision-makers are handed a probability with no notion of what to *do* with it. DisasterMind addresses the full chain — **sense → predict → decide → disseminate** — and, critically, treats *validation and honesty* as first-class engineering, so its outputs can earn the trust an evacuation decision requires.

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3. System architecture

DisasterMind is a **three-tier multi-agent system** communicating over a single message bus. A hazard signal flows up the tiers and back down to dispatch:

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  Tier 3 · SENSE           Tier 2 · THINK                                 Tier 1 · DECIDE      Tier 3
· ACT

────────────────────────────────────────────────────────────────────────────────────────────────────
  raw_feed                                                                commander
  iot_telemetry  ──▶  prediction ─▶ cascade ─▶ resource_plan ─▶ routing_plan ─▶ field_order ─▶ review ──▶
dispatch
                                                                                         └───▶

escalation (human)
```

- **Tier 3 (sensing & acting):** ingestion adapters for each hazard, IoT gateways, social-signal intake, and outbound dispatch.
- **Tier 2 (analysis & planning):** per-hazard prediction agents, cascading-effect models, resource optimisation, evacuation routing, and field tasking.
- **Tier 1 (command):** the commander agent, escalation logic, and concurrent multi-incident orchestration.

Message bus & topics. All agents publish/subscribe over an in-memory `MessageBus` on typed topics (`raw_feed` , `iot_telemetry` , `prediction` , `cascade` , `resource_plan` , `routing_plan` , `field_order` , `commander_review` , `escalation` , `dispatch`), with five priority levels (CRITICAL→INFO). A coordination loop ticks on a fixed cadence, builds the agent DAG, and routes every message through all stages.

Hazard modules. Three domains run as independent pipelines: **A — cyclone/ flood**, **B — earthquake**, **C — fire/structural collapse** (plus cross-module coordination).

Graceful degradation. The core runtime is standard-library only. Heavy capabilities (gradient-boosted models, OR-Tools, Postgres/Timescale, Kafka, satellite feeds) are optional; when absent, the system falls back to deterministic in-process implementations rather than failing.

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4. Capabilities in detail

4.1 Multi-hazard risk prediction

Per-hazard models score per-asset/per-cell risk from physically meaningful drivers, each with explainable feature attributions (SHAP-style):

- **Earthquake** — magnitude, depth, location physics → measured damage-grade outcome (instrumental intensity / loss alert).
- **Cyclone/flood** — rainfall accumulations, river discharge and trend, surge, seasonality → flood-threshold exceedance.
- **Fire** — fire-weather (temperature, humidity, wind), drought/dryness streaks, seasonality → next-day ignition.

Each model runs the validated, dependency-free calibrated baseline by default and will use a trained gradient-boosted backend when the optional ML extra is installed; either path returns calibrated probabilities, so the system behaves identically — only more accurately — as capabilities are added.

4.2 Cascading-effect modeling

Beyond first-order risk, the platform models downstream consequences: earthquake **aftershock sequences** (Omori-law decay), **flood propagation** along river networks, and fire spread — so the coordinator anticipates the second wave, not just the initial event.

4.3 Resource allocation & evacuation routing

- **Resource optimiser** allocates scarce assets (teams, vehicles, supplies) across competing demands.
- **Road-network routing** plans evacuation routes over real OpenStreetMap road graphs to real shelter locations, with road-closure awareness.

4.4 Evacuation decision layer

The component that turns a forecast into an actionable order:

- **Clearance-time model** — given zone population, road egress capacity, and shelters, computes how long the zone takes to empty and *by when the order must be issued*; flags zones that cannot be cleared in the available lead time and recommends vertical (in-place) evacuation; supports contraflow and egress sensitivity analysis.
- **Vulnerability-weighted phased evacuation** — schedules cohorts (hospitalised, elderly/mobility-impaired, transport-dependent, general public) against scarce assisted-transport capacity and surfaces exactly which group is left behind when warning time runs short.
- **Dissemination & warning-response modeling** — multi-channel reach (cell broadcast, SMS, siren, radio, community wardens) combined with a *compliance* model that consumes both lead time and false-alarm rate (capturing the cry-wolf effect), reporting who is left behind and *why* (not reached vs. won't comply vs. cannot be moved).
- **Cost/benefit & break-even** — net lives saved, evacuation-induced casualties, and the probability threshold below which ordering an evacuation costs more than it saves (over-evacuation guard).
- **Integration capstone** — one per-zone recommendation (`ORDER_BY_DEADLINE` / `NOT_CLEARABLE_VERTICAL` / `BELOW_BREAKEVEN_HOLD` / `NO_ACTIONABLE_WARNING`) plus a defensible decision record ("what we knew, when, and what we recommended").

4.5 Human-in-the-loop command & escalation

The commander tier reviews agent recommendations, manages an escalation queue with approve/reject/timeout semantics, generates escalation memos and situation briefings, and runs concurrent incidents (one coordination DAG per active disaster). Every consequential action crosses a defined authority threshold and requires human approval — mass evacuations (> 10,000 people), cross-jurisdiction resources, and military assets escalate for sign-off (auto-executing only on timeout), while declaring a state of emergency, deploying armed forces, international aid, and critical-infrastructure requisition are *human-only* and never act without a commander. Only routine field tasking dispatches unattended. The system never issues a mass-evacuation order autonomously; it recommends, and a human acts.

4.6 Field operations

Device-facing field contracts and a client that close the dispatch → acknowledge → GPS-update loop, with a durable outbox (terrestrial first, satellite fallback, offline queue) for connectivity-denied environments.

4.7 Emergency alerting

Standards-compliant **CAP 1.2** emergency-broadcast XML generation for interoperability with public warning systems.

4.8 Validation & evidence framework *(the platform's signature capability)*

A complete, dependency-free evaluation suite that scores and documents every model on real, held-out data:

- **Metrics** — rank-based ROC-AUC, Brier score, accuracy, calibration/ECE.
- **Decision-point metrics** — POD (detection), FAR (false-alarm ratio), CSI, frequency bias, HSS at a chosen operating threshold, with explicit miss-vs-false-alarm cost accounting.

- **Operational-baseline comparison with significance** — paired-bootstrap tests (p-values, confidence intervals) against the incumbents a forecaster uses today.
- **Lead-time-vs-POD curves** — the *actionable warning time*, not just accuracy.
- **Blocked cross-validation** — leave-one-region-out and rolling-origin, reporting the worst block, not just the average.
- **Calibrated uncertainty** — isotonic recalibration and split-conformal prediction sets with verified coverage.
- **Fairness audit + remediation** — per-subgroup detection rates, with equalized-odds remediation that states the false-alarm *cost of equity* and classifies residual gaps by cause.
- **Rare-severe-event (tail) analysis** with bootstrap confidence intervals.
- **Drift detection & retraining triggers** — PSI/KS feature drift and a skill-decay-driven retrain decision.
- **Degraded-input robustness** — graceful-degradation curves under sensor failure.
- **Shadow mode** — an append-only, hash-chained journal that records live predictions before outcomes are known, scores a season against reality, and exports a complete record for independent review.
- **Backend benchmark** — the optional gradient-boosted backend is scored against the default baseline on identical splits and held to a snapshot in CI, so any accuracy claim for it is a reproducible measurement.
- **Physics-model validation** — the analytic models (HAZUS-style fragility, Omori-Utsu aftershock decay, cellular-automata fire spread) are checked against the published invariants of their source literature.

4.9 Historical hindcasting & backtesting

- **Named-event replay** — Cyclone **Fani (2019)** and **Amphan (2020)** are replayed from their real best-tracks using only pre-cutoff data (leak-free), driving the full activation and coordination pipeline and scoring against the documented outcome (people evacuated, fatalities, landfall intensity).
- **Population-scale cyclone backtest** — the entire modern North-Indian-Ocean record: **92 named, India-landfalling cyclones (1990–2025)** from NOAA IBTrACS, scored for activation lead time and landfall-extrapolation error.
- **Full-pipeline backtest** — forecast → evacuation decision → scored against reality, on real storms.

4.10 Live data ingestion

Adapters target authoritative live sources: **USGS** and **India NCS** seismology, **Open-Meteo** (GloFAS flood + weather), **NASA FIRMS** active-fire detections, **India-WRIS** river monitoring, **ISRO Bhuvan** flood inundation, and **IMD** forecasts. Several operate live and key-free today; the remainder are wired and require provider keys/endpoint configuration.

4.11 Production engineering

Health/readiness probes, retry and circuit-breaker resilience, graceful shutdown; API authentication, rate limiting, payload validation, CORS hardening; metrics collection with Prometheus exposition; distributed tracing with per-incident latency; a tamper-evident audit chain with signing, retention, and backup; durable storage (Postgres/Timescale, Elasticsearch, MinIO) with versioned migrations; cross-district mutual-aid federation; a structured-logging stack; and a system self-check ("doctor") plus a load/throughput benchmark harness.

4.12 Web command console (`clients/web`)

A unified Vite + React 19 + TypeScript application (MapLibre GL maps, Recharts) with four modules — **Commander Dashboard**, **Escalation**, **Field Ops**, and **Post-Incident Report** — talking to the platform API (configurable base URL), with live status, override controls, SHAP explanations, and PDF report export.

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5. Validation results (real data, leak-free, out-of-sample)

All figures are produced by the validation suite on committed real-data fixtures with strictly temporal splits; operating thresholds and calibrators are fit on a calibration split, never on the test set. See [`docs/TECHNICAL_REPORT.md`](docs/TECHNICAL_REPORT.md) for the full evaluation including baseline-significance tables, worst-block generalisation, and a frank failure analysis; reproduce every number with `make reproduce` .

> ***Reproducibility and model design.*** *The figures below are produced by the platform's default model — a deterministic, calibrated logistic fit that uses only the standard library. Because it is fixed-seed and dependency-free,* `make reproduce` *regenerates every published value exactly from the committed fixtures (Δ = 0), so the table is a continuously verified artefact rather than a static claim. An optional gradient-boosted backend (XGBoost) is benchmarked on the identical splits and calibration by* `make compare-backends` *: it improves earthquake ranking (AUC 0.937 → 0.950) and matches the baseline on flood and fire, so it is offered as a hazard-specific, opt-in upgrade while the dependency-free model remains the shipped default. Flood risk is modelled from tabular hydro-meteorological drivers.*

Hazard	Data source	Out-of-sample AUC	Brier	ECE	Actionable lead (POD ≥ 80%)
Earthquake	USGS catalog (2013–2017, M4.5+)	**0.937**	0.011	0.002	n/a (instantaneous)
Flood	GloFAS-ERA5, 12 Indian basins (2010–2023)	**0.944**	0.028	0.004	**168 h (7 days)**

Fire (PNW)	USDA FPA-FOD + ERA5 (2012–2018)	**0.837**	0.121	0.023	**72 h (3 days)**
Fire (India)	NASA FIRMS VIIRS + ERA5 (2015–2024)	**0.855**	0.153	0.015	seasonal*

Beats the operational incumbents, with statistical significance:

- **Flood** beats *persistence* (the standard no-model hydrological forecast, $p \approx 0.004$) and *seasonal climatology* ($p \approx 0.004$).
- **Fire** beats the *Angström fire-danger index* — both on US ($p \approx 0.004$) and on real Indian data ($p \approx 0.02$).
- **Earthquake** statistically matches a GMPE ground-motion attenuation baseline on the damage label and **beats USGS PAGER by +0.22 AUC** on the felt-report label.

Population-scale cyclone evidence (92 real storms, IBTrACS): the system would have raised a cyclone alert a **median of 54 hours before landfall**, with **≥48 h lead for ~58%** of storms and ≥72 h for ~40%.

Operational decision quality is reported per hazard at the dispatch threshold (POD/FAR/CSI), with calibration repaired by isotonic recalibration (e.g., earthquake ECE 0.21 → 0.002), conformal coverage at target, blocked cross- validation worst-block scores, a published fairness audit with remediation, and rare-severe tail analysis.

* The India fire model is trained on the 2015–2021 FIRMS fire seasons and tested out-of-sample on three held-out seasons (2022–2024). At the dispatch threshold it reaches POD 0.92 / FAR 0.37 (CSI 0.60) — the strongest operational decision quality of the four models. "Seasonal" lead reflects the next-day ignition framing.

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6. Data sources (committed, reproducible)

Domain	Source	Coverage
Earthquakes	USGS FDSN catalog	~36,500 real M4.5+ events, 2013–2017
Floods	GloFAS-ERA5 discharge + ERA5 rainfall (Open-Meteo)	12 Indian river-basin sites, 2010–2023
Fire (US)	USDA FPA-FOD occurrences + ERA5	12 Pacific-Northwest cells, 2012–2018
Fire (India)	NASA FIRMS VIIRS detections + ERA5	10 Indian fire-belt cells, 2015–2024
Cyclones	NOAA IBTrACS best tracks	92 named India-landfalling storms, 1990–2025
Cyclone cases	IBTrACS + documented outcomes (IMD/EM-DAT)	Fani 2019, Amphan 2020
Infrastructure	OpenStreetMap road graph, Copernicus DEM, Census	Puri / Fani impact zone
External outcomes	GDACS (UN/EC) declared disasters	166 Indian flood/cyclone events

All data is real and citable; every fixture records its provenance. The validation suite runs fully offline against committed fixtures; a separate fetch utility rebuilds them from the free public APIs.

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7. Technology & deployment

- **Language/runtime:** Python ≥ 3.11, standard-library-first; optional extras for ML (XGBoost/scikit-learn/SHAP), optimisation (OR-Tools/PuLP), geospatial, messaging (Kafka), storage (Postgres/Elasticsearch/MinIO), and feeds.
- **API:** framework-free dashboard service over HTTP + WebSocket (health/ readiness probes, topics, incidents, escalations with approve/reject, live stream), OpenAPI-specified.
- **Frontend:** Vite + React 19 + TypeScript, deployable to Vercel.
- **Packaging/CI:** multi-stage Docker image (pinned base, non-root, healthcheck); GitHub Actions pipeline with a Python test matrix, lint, type-check, security scanning (Trivy + pip-audit), and a CycloneDX SBOM.
- **Deployment targets:** container/Railway and Kubernetes manifests with a SQL schema.
- **CLI:** `run`, `simulate`, `train`, `eval`, `doctor`, `serve`, `verify-audit`, plus dedicated entry points for validation, hindcasting, diagnostics, and the demo.

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8. Maturity & roadmap

Current status — a validated research-grade platform. The prediction and evaluation layers are rigorous and reproducible; the coordination and evacuation- decision layers are complete and operate end-to-end on real historical events.

Known limitations (stated transparently):

- Earthquakes cannot be *forecast* on an evacuation horizon; the earthquake module performs rapid impact assessment, and the evacuation framing applies to cyclone/flood/fire.

- Evacuation-planning parameters (clearance times, compliance rates, casualty rates) are explicit, tunable planning assumptions pending calibration against agency ground truth.
- Some outcome labels are well-justified proxies (discharge exceedance, instrumental intensity, satellite detections) rather than surveyed losses.
- Regional generalisation is weaker than the headline: worst-block AUC falls to ~0.80 for both fire models (see the technical report's failure analysis).

Roadmap to operational deployment:

1. Calibrate the evacuation layer against district-level historical response data.
2. Complete live-feed integration (provider keys + endpoint configuration) and expand the India fire record to multiple years.
3. Run a full **shadow-mode season** in partnership with a state disaster- management authority — predicting live, acting on nothing, scored against actual outcomes — followed by independent review. This is the decisive step toward trusted operational use.

DisasterMind — earlier, better-targeted, and more equitable disaster warnings, built to earn the trust an evacuation decision requires.